

Comptia A+ Assignment

[A+ - Understanding and maintenance of Networks]

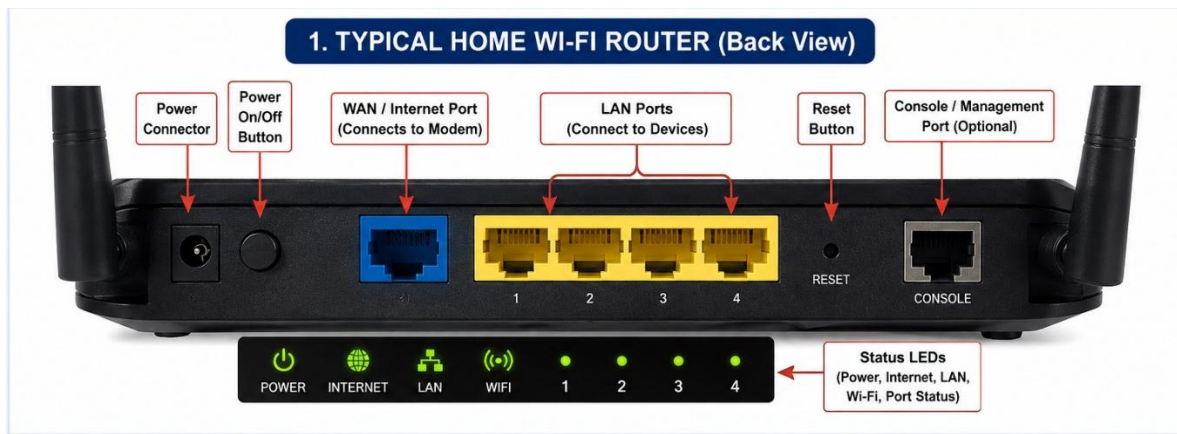
Routers and Switches

Basic functions and differences between routers and switches. Identify ports and LEDs on a router and switch.

Task

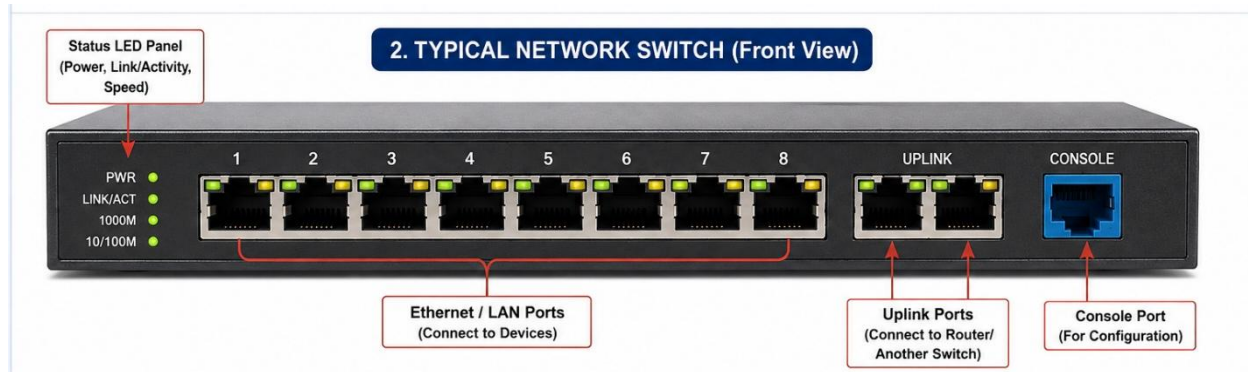
- ❖ **1. Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.**
 - A router connects the local network to other networks such as the Internet, while a network switch connects multiple wired devices within the same local network. Both devices have different ports and status LEDs that help identify connections and network activity.
 - 1. Typical Home Wi-Fi Router:
 - A typical home Wi-Fi router may have the following ports and indicators:
 - **WAN/Internet Port:** Connects the router to the modem or Internet service.
 - **LAN Ports:** Connect wired devices such as PCs, printers, or switches.

- **Console/Management Port:** Available on some routers for configuration and troubleshooting.
- **Status LEDs:** Usually indicate power, Internet/WAN status, LAN activity, and Wi-Fi activity.



➤ 2. Typical Network Switch:

- A network switch is mainly used to connect multiple wired devices within a LAN.
- **LAN/Ethernet Ports:** Connect computers, printers, access points, servers, or other network devices.
- **Console Port:** Found on many managed switches and used for configuration and troubleshooting.
- **Uplink Port:** Used to connect the switch to another switch, router, or network device.
- **Status LEDs:** Show power, link/activity, and sometimes the speed/status of individual ports.



❖ **2. List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic. Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.**

➤ Given the difference between routers and switches:

Difference	Router	Switch
Main Function	Connects different networks, such as a home LAN to the Internet.	Connects multiple devices within the same local network (LAN).
Address Used	Uses IP addresses to determine where network packets should be forwarded.	Uses MAC addresses to forward Ethernet frames to the correct device.
Traffic Handling	Routes traffic between different networks/subnets and selects the appropriate network path.	Forwards traffic within a LAN to the specific connected device/port.

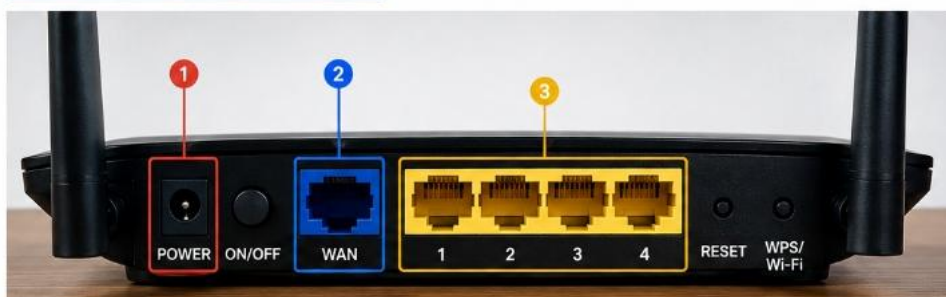
❖ **3. Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.**

➤ Router and Switch – Ports and LED Identification:

➤ 1. Wi-Fi Router

- A typical Wi-Fi router has different ports and LED indicators for power, Internet connection, and wired network connections.
- **Power Port:** Connects the router to its power adapter.
- **WAN/Internet Port:** Connects the router to the modem or Internet connection.
- **LAN Ports:** Used to connect computers, printers, switches, and other wired devices.
- **LED Indicators:** Show the power status, Internet connection, LAN activity, and Wi-Fi status.

1. WI-FI ROUTER (Back View)



- 1 Power Port**
Connects the router to the power adapter.

2 WAN Port
Connects to modem / Internet.

3 LAN Ports (1,2,3,4)
Connect to computers, TV, printers, etc.

LED INDICATORS (Router)			
LED	Color	Purpose	
	Power	Green	Indicates router is ON.
	Internet	Green/Blue	Shows Internet connection status.
	LAN	Green	Indicates LAN port connection/activity.
	Wi-Fi	Green	Indicates Wi-Fi is ON.

➤ 2. Network Switch:

- A network switch connects multiple wired devices within a local network and uses LEDs to show the status of its connections.
- **Power Port:** Connects the switch to its power adapter.
- **Ethernet/LAN Ports:** Used to connect computers, printers, access points, and other network devices.
- **Uplink Port:** Used to connect the switch to a router or another switch.
- **LED Indicators:** Show power, link/activity, and sometimes the connection speed of each port.

2. NETWORK SWITCH (Front View)



- 1 Power Port**
Connects the switch to the power adapter.
- 2 Ethernet Ports (1-8)**
Connect to computers, printers, access points, or other network devices.
- 4 Uplink Port**
Connects to router or another switch.
- 3 Reset Button**
Resets the switch to factory default settings.

LED INDICATORS (Switch)		
LED	Color	Purpose
Power	Green	Indicates switch is ON.
Link/Act	Green (On) Blinking	On = Link established Blinking = Data activity
Speed	Green	On = 1000 Mbps Off = 10/100 Mbps

❖ 4. Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

➤ Gaming Café Network Setup:

➤ Scenario:

- A gaming café needs a reliable network for IPL streaming and multiplayer gaming. The café has 8 gaming PCs that need stable Internet access and communication with the gaming network.

➤ Network Setup:

➤ 1.Connect the ISP line to the Router first.

- The ISP/Internet connection is connected to the WAN/Internet port of the router. The router manages the Internet connection and routes traffic between the café's local network and the Internet.

➤ 2.Connect the Router to the Switch:

- One LAN port of the router is connected to an Ethernet port on the network switch using an Ethernet cable.

➤ 3.Connect all 8 Gaming PCs to the Switch.

- Each gaming PC is connected to a separate Ethernet port on the switch using Ethernet cables.

➤ Simple Connection Diagram

- **ISP/Internet → Router → Switch → 8 Gaming PCs**

➤ Justification:

- I would connect the router first because it acts as the gateway between the ISP/Internet and the café's local network. I would then use a network switch to provide enough Ethernet ports for all 8 gaming PCs. This setup gives the PCs a stable wired connection, which is suitable for multiplayer gaming and smooth IPL streaming.

➤ Final Answer:

- Router → Switch → 8 Gaming PCs is the preferred setup because the router handles communication with the Internet, while the switch efficiently connects multiple gaming PCs within the local network.